

EEF Meeting Minutes, Thursday Nov 10, 2022

Erlang Ecosystem Foundation, Minutes of a Regular Meeting of the Board of Directors Meeting #73, November 10, 2022

A regular meeting of the Erlang Ecosystem Foundation was held over group conference call (Zoom). The meeting began at 16:04 UTC.

Attendance

The following members of the board of directors were present (8/11): Sophie DeBenedetto, Ben Marx, Bryan Paxton, Kenneth Lundin, Miriam Pena Villanueva (joined at :08 after item 1), Peer Stritzinger, Sebastian Strollo, and Alistair Woodman.

The following members of the board of directors were absent: Francesco Cesarini, Maxim Fedorov, Frédéric Trottier-Hébert

Agenda Items

1. Approval of minutes from previous meetings.

RESOLVED, that the Erlang Ecosystem Foundation approve the minutes as presented as representing a true and accurate record of the board meeting #72 held on October 27, 2022. Unanimously approved.

2. Intermediate actions and activities: Annual General Meeting, AGM, held over Zoom on Nov 3 and Nov 4. Marcom team is setting up a Mastodon account.

3. No treasury and membership update this meeting.

4. Unanimous consent votes: none scheduled

5. Stipend applications:

1. Static Type Checker, from Annette Bieniusa at TU Kaiserslautern. For \$25,000 to support development. The OTP team has had time to review and now brought to a vote in this meeting (offline vote via email didn't get a response from everyone, but only positive responses received).

RESOLVED, that the Erlang Ecosystem Foundation approve the stipend application from Annette Bieniusa and the Technical University of Kaiserslautern \$25,000 to support the development of a new type checker. Unanimously approved.

2. Application from Tanguy Losseau for \$950 for the initial step to integrate multiple Basic Linear Algebra Subprograms (BLAS) libraries as NIF:s. Peer is sherpa.

RESOLVED, that the Erlang Ecosystem Foundation approve the stipend application from Tanguy Losseau for \$950 to support the first step in the development of a new BLAS library as a NIF. Unanimously approved

6. Report from Code BEAM America: EEF sponsored 30 students. Positive experience, mix of in person and online attendance. Sophie and Miriam were there.
7. Report from Annual General Meeting: Fewer attendees than last year, still very much appreciated. For many the testimonials were the highlight.
8. Any other business:
 1. Bryan reported that the jobs board has kicked off (with staff from Erlang Solutions). Bryan is helping getting the project off the ground.
 2. Bryan has started working on the second half of the blog project (by him self).
9. Alistair declared the meeting adjourned at 17:00 UTC.

Dated November 10, 2022

Secretary of the meeting: _____ Sebastian Strollo

Attachments:

1. Stipend request from Annette Bienusa, TU Kaiserslautern (2 pages)
2. Stipend request from Tanguy Losseau (2 pages)

Stipend Type : Development Work

First name : Annette

Last Name : Bieniusa

Organization Name : TU Kaiserslautern

Country : Germany

Twitter : @anne_biene

Linkedin : <https://www.linkedin.com/in/annette-bieniusa-b0807>

Website : <https://softech.informatik.uni-kl.de>

Intended product of project :

Set-theoretic types and semantic sub-typing fit Erlang's feature set very well. They allow expressing nearly all constructs of its type language and provide means for statically checking type signatures. In our work, we bring set-theoretic types to Erlang and demonstrates how existing Erlang code can be statically type-checked without or with only minor modifications to the code. At the end of the project, we want to deliver a static analyzer that is capable of identifying erroneous code in large-scale Erlang code bases.

Purpose

There is a growing number of static type checkers for Erlang, including the well-established dialyzer, gradualizer, and more recently equalizer. However, most of them have short comings - either due to the underlying theory or due to issues with the implementations. We want to provide another industry-grade static type checker that hits the sweet spot of providing strong guarantees and few restrictions / requirements for change of existing codebases while covering the language semantics. We believe that the Erlang language and community will benefit from the results and a more rigorous alternative to dialyzer's success typing approach.

Development Plan

We have already implemented a prototype version of the type checker to validate the usage of set-theoretic types for the Erlang language. (As we have a publication with the first results in submission, we can only give confidential access at the moment to the code base for evaluation of the proposal, similarly for the paper.) To make the typechecker usable in practice, we need to address the following aspects: - coverage of all Erlang language features - implementation of a type check manager to allow for composable and incremental type checks per module - improvement of error messages to guide the programmer in finding the actual issue - proposals for refinements / fixes for the type annotations in the standard libraries The work is done by Albert Schimpf and Annette Bieniusa from TU Kaiserslautern (Germany) and Stefan Wehr (HS Offenburg, Germany). We are further planning to involve student programmers to support us with the software development. We expect the implementation work to cover approx. 4-5 man months which we would like to spread over 6 months.

Dissemination : We plan to summarize and document our work on a project webpage and present them in scientific and industrial conferences.

Delivery Method : The code will be made available on Github under the Apache or MIT license for free use in the community.

Stipend Amount (USD) : 25000

Stipend Type : Development Work
First name : Tanguy
Last Name : Losseau
Country : Belgium

Intended product of project :

This project intends to write an erlang BLAS library supporting multiple implementation (cblas, openblas at least), the Basic Linear Algebra Subprogram being a set of heavily optimised functions written in Fortran, C and usable through erlang's NIFS.

Purpose

Currently, Erlang does not have a complete matrix library. An open source incomplete library exists (<https://github.com/sklassen/erlang-linalg-native>), but substantial speedups can be obtained using BLAS in NIFS (see <https://dial.uclouvain.be/memoire/ucl/fr/object/thesis%3A33858>). Multiple projects have done this, with limitations: not all BLAS functions can be used, through modified interfaces not always documented. This project aims to produce a wrapper safely exposing the BLAS api as it is, making the usage of matrices and vector easy and efficient in Erlang.

Development Plan

This project is expected to take around 8 days of work, and I should be done with it before mid october. Follow up projects include a LAPACK wrapper (library for matrix inversion, various problem solvers), and a matrix library.

Dissemination : Results will be announced on the Erlang forums.

Delivery Method : The resulting library will be available on a github repo.

Stipend Amount (USD) : 950

BLAS interfaces

Two blas wrappers have already been written for Erlang:

- Numerl from <https://github.com/CoupletB/numerl>
- Blas from <https://github.com/dgud/blas>.

Dgud's BLAS

This implementation is divided in two erlang files:

- Blas.erl, which provides high level vector/matrix records and matching simplified BLAS functions
- Blasd_raw.erl, which provides a tighter BLAS interface to most BLAS functions.

All BLAS functions operate on a nif resource, **avec_r**, which is just a C double pointer:

```
typedef struct {  
    double v[1];  
} Avec;
```

Each BLAS function has two interface:

- A complete version to which is added a pointer shift: `scal(N, Alpha, X, XStart, Xinc)`

- A basic version: `scal(Alpha, X)`

The basic version expands the nif resource arguments with no pointer shift/stride of 1: `scal(Alpha, X) -> N = cont_size(X), scal(N, Alpha, X, 0, 1)`.

Each BLAS has various simplified interface; for example copy has 4 different wrappers.

As of scheduling, each BLAS runs in a dirty NIF; `enif_consume_timeslice` is used indicating a naïve time complexity, the number of elements iterated on.

CoupletB's Numerl

This BLAS wrapper is the backend of the project Erlblas (<https://github.com/zelirio/ErlBlas>). Both high level (add, sub...) and direct BLAS functions are exposed in a single file, `numerl.erl`.

The mutable data container is Resource Matrix (int number_of_col, int number_of_rows, resource doubleArray) where doubleArray is just a c double pointer.

The few BLAS implemented provide Dgud's basic version: for example, `scal(Alpha, X)` with no stride.

Unlike Dgud, no interface is provided to shift mutable arrays.

Finally, `enif_consume_timeslice` is not used, which can only make concurrent scheduling harder.